# Qijia Shao

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## Research Interests

My research lies at the intersection of **Mobile & Ubiquitous Computing**, **Human-Computer Interaction** and **Artificial Intelligence**. We develop novel sensing rationales and human-centered AI systems to make previously inaccessible physical and physiological signals measurable, interpretable, and actionable. My work addresses two complementary challenges: opening new sensing channels when the body, environment, or activity makes conventional sensing impossible, and closing sensing–intervention loops by translating sensed information into timely and appropriate support with AI while preserving human agency. These systems enable applications in healthcare, sports, education, and everyday interaction. My research is inherently interdisciplinary, drawing inspiration from clinical and health sciences, cognitive science, materials science, signal processing, and electrical engineering.

## Appointment

2024.09-Now **The Hong Kong University of Science and Technology**
Division of Integrative Systems and Design (ISD)
Department of Computer Science and Engineering (CSE)
**Assistant Professor**

## Education

2018-2024 **Columbia University**
Ph.D. in Computer Science
Advisors: **Prof. Xia Zhou and Prof. Fred Jiang**

2018-2022 **Dartmouth College**
Master of Science in Computer Science (Conferred during Ph.D. program)
Transferred to Columbia University
Advisors: **Prof. Xia Zhou and Prof. Devin Balkcom**

2014-2018 **University of Electronic Science and Technology of China**
Bachelor of Engineering in Computer Engineering
Yingcai Honours College (for top 5%)
GPA: **3.99/4.0**, Avg.Score: **91.3/100**

## Selected Honors & Awards

2026 Hong Kong RGC Early Career Award
2026 ACM SIGMOBILE Research Highlights
2026 ACM HEALTH Distinguished Reviewer Board
2025 IEEE Pervasive Computing Emerging Rockstar
2024 ACM UbiComp Gaetano Borriello Award Finalist
2024 ACM MobiSys Best Paper Award
2024 ACM MobiSys People's Choice Best Demo Award
2024 NSF Rising Stars in Cyber-Physical Systems

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| 2024 | ACM MobiSys Rising Stars |
| 2023 | ACM MobiCom Best Demo Award |
| 2023 | ACM UbiComp Best Teaser Award |
| 2022 | Grand Prize at Dartmouth Innovation and Technology Festival |
| 2020 | ACM HotMobile 2020 Best Demo Award |
| 2019 | NSF Awesome Discoveries |
| 2018 | Dartmouth Fellowship |
| 2018 | Excellent Undergraduate Student at UESTC |
| 2018 | Outstanding Undergraduate Thesis Award at UESTC |
| 2016, 2017 | National Scholarship, by the Ministry of Education of China |

## Industry Experience

Summer 2023 **Research Intern**, *Samsung Research America, Mountain View*
**Mentors**: Dr. Li Zhu and Dr. Jilong Kuang
- Led a project on mitigating the cross-user performance variation of rPPG-based SpO2 estimation.
- Awarded a Samsung A1 patent.
- The proposed method is **being deployed to Samsung products** (phones and TVs).
- Published a **ICASSP'24** paper.

Summer 2022 **Research Intern**, *Snap Research, NYC*
**Mentors**: Dr. Jian Wang and Dr. Shree Nayar
- Led a project on reducing the motion-to-photon latency for AR/VR.
- The proposed technique is on the roadmap for Snapchat App.
- Published the N-euro Predictor paper (**UbiComp'23**).

Summer 2021 **Research Intern**, *Signify (Philips Research), Remote*
**Mentor**: Dr. Jin Yu
- Led a project on sensor data processing.
- Built a scalable and robust probabilistic model and implemented the whole pipeline.
- Improved the system performance by 19% and filed a patent.

## Professional Services

Editorial Board: **Associate Editor of the Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies (IMWUT/UbiComp)**, *2024-now*

Technical Program Committee:
**ACM MobiSys**, 2025, 2026
**ACM SenSys**, 2025, 2027
**ACM MobiCom**, 2027

Organizing Committee:
**Gadget Show Chair**, UbiComp 2026
**Local Chair**, MobiCom 2025
**General Chair**, MobiCom-SmartWear 2025
**Web Chair**, MobiSys 2021

Reviewers:
**ACM UbiComp/IMWUT**, *2020-2026*
**ACM CHI**, *2020, 2023, 2024, 2025, 2026*
**ACM MobiCom**, *2024*
**ACM UIST**, *2024*, 2025 (*special recognition for outstanding reviews)*
**ACM MobileHCI**, *2024*
**IEEE ICRA**, *2023, 2025*
**ACM Transactions on Computing for Healthcare**, *2025, 2026*
**IEEE Internet of Things Journal**, *2025, 2026*
**ACM Transactions on Internet of Things**, *2024, 2025, 2026*

**IEEE Transactions on Mobile Computing**, *2023, 2025, 2026*

Others **Session Chair**, *ACM MobiCom 2025, UbiComp 2024*

**Keynote Speaker**, *ACM UbiComp-MIMSVAI 2024*

**JEDI (Justice, Equity, Diversity, and Inclusion) Ambassadors**, *ACM UbiComp 2023*